

Customer Churn Analysis for Telecom Industry

Project Report | Data Analytics Internship

1. Introduction

Customer churn — when a subscriber discontinues service — is one of the most critical challenges facing telecom companies today, where market competition is intense and the cost of acquiring a new customer far exceeds the cost of retaining an existing one. This project analyzes customer data from a telecom provider to understand the factors driving churn and to build a predictive system that flags at-risk customers before they leave.

2. Abstract

This project uses the Telco Customer Churn dataset (7,032 customers, 21 attributes) to predict customer churn using machine learning. SQL-based aggregation was used to analyze churn patterns across contract types, tenure groups, and payment methods. A binary classification model (Random Forest, benchmarked against Logistic Regression) was built to predict churn, achieving an ROC-AUC score of 0.84. Model explainability was performed using permutation feature importance to identify the key drivers of churn. Customers were further segmented into three actionable groups — At Risk, Loyal, and Dormant — to support targeted retention strategies. The project concludes with data-driven recommendations that could help reduce churn by an estimated 5-10%.

3. Tools Used

- Python — Pandas, NumPy for data cleaning and manipulation
- SQLite (SQL) — data aggregation and churn-rate queries
- Scikit-learn — Logistic Regression and Random Forest classification models
- Matplotlib, Seaborn — exploratory data analysis and visualizations
- Permutation Importance (scikit-learn) — model explainability
- Google Colab — development environment
- python-pptx, ReportLab — automated report generation (PPT & PDF)

4. Steps Involved in Building the Project

1. Data Cleaning: Handled missing/invalid TotalCharges values, removed duplicates and irrelevant identifiers
2. SQL Aggregation: Queried churn rate by contract type, tenure group, and payment method using SQLite
3. Exploratory Data Analysis: Visualized churn patterns across demographics, services, and billing attributes
4. Feature Engineering: Encoded categorical variables (binary + one-hot encoding) for model readiness
5. Model Building: Trained and compared Logistic Regression and Random Forest classifiers
6. Model Evaluation: Assessed performance using confusion matrix, classification report, and ROC-AUC
7. Explainability: Applied permutation importance to identify top churn-driving features
8. Customer Segmentation: Classified customers into At Risk, Loyal, and Dormant segments based on churn probability and tenure
9. Reporting: Compiled findings into a PowerPoint presentation and this PDF summary report

5. Conclusion

The analysis confirms that contract type, tenure, and monthly charges are the strongest predictors of churn, with month-to-month customers on high bills and short tenure being the most likely to leave. The Random Forest model achieved reliable predictive performance (ROC-AUC: 0.84), and the resulting customer segments give the business a clear, actionable framework: retain 'At Risk' customers through proactive

offers, reward 'Loyal' customers, and re-engage 'Dormant' ones. Implementing these targeted strategies can meaningfully reduce churn and protect recurring revenue.